

Research Statement

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My research interests lie at the intersection of innovation, empirical corporate finance, and mergers & acquisitions. I combine large-scale textual analysis, patent data, and financial markets data with modern machine learning methods to study how information, technology, and organizational capabilities shape firm behavior and value creation. My dissertation papers focus on two parts. The first part examines technological complexity as a fundamental characteristic of innovation and show that complexity has important implications for corporate decisions and market valuations.

Job Market Paper

1 Complex Technologies and Economic Value

Technological innovation is a fundamental driver of long-run economic growth, yet innovations differ substantially in the complexity of the knowledge they combine. Some technologies rely on relatively standardized and easily replicated components, while others require the integration of diverse and highly interconnected knowledge. Despite the importance of this distinction, technological complexity remains difficult to quantify consistently, particularly in a way that can be linked to industry characteristics and financial outcomes. Existing measures generally capture only selected dimensions of complexity, and there is limited evidence on how technological complexity evolves across industries, what drives it, and whether financial markets place value on it.

This paper develops a novel industry-level measure of technological complexity from U.S. patent data and introduces a two-step probabilistic concordance that maps CPC-based measures into successive four-digit NAICS classifications. This concordance enables patent-based technological complexity to be linked consistently with industry-level economic and financial data over time. The proposed measure performs favorably relative to alternative complexity metrics by capturing multiple dimensions of structural heterogeneity simultaneously. Using this measure, I document several new patterns in the evolution and determinants of technological complexity. Complexity generally increases over time, consistent with the cumulative nature of innovation, with industries progressively combining a broader and more interconnected set of knowledge components. This trend continues until the COVID-19 pandemic, after which complexity declines substantially in several labor-intensive industries while remaining comparatively resilient in high-technology industries. Because the level of the complexity score differs systematically across industries, I construct an industry-demeaned “excess” complexity measure to facilitate comparisons across industries.

I show that the determinants depend importantly on product-market competition. In more competitive industries, technological complexity is associated with stronger innovation incentives and greater financial flexibility. Less profitable firms may have stronger incentives to develop differentiated technologies, while lower reliance on debt allows firms to sustain the investment and experimentation required to develop complex technologies. In more concentrated industries, by contrast, firms face weaker competitive pressure to continually expand their technological capabilities, and the relationship between financial characteristics and complexity becomes substantially weaker.

Finally, I show that technological complexity is economically meaningful for valuation. Industries characterized by greater technological complexity tend to produce patents with higher measured value, suggesting that financial markets reward innovations that combine more sophisticated knowledge structures. Stock-market responses are also positively related to complexity over a substantial range, but this relationship weakens at very high levels of complexity. One interpretation is that investors value the innovative potential associated with complex technologies but become more uncertain about commercialization costs, implementation difficulties, and future profitability when technological systems become excessively complex.

Other Working paper

1 The LLM Oracle Fallacy in Finance (with Debarshi Nandy and Shubhramshu Shekhar)

This paper is the first of my two papers on rationales of mergers. The paper is a collaboration with my advisors who introduced me to combining Large language models (LLMs) with finance studies. When the analysis began in early-2026, LLMs were widely used in finance research to measure economic constructs at scale. Researchers feed earnings calls, merger documents, and central bank communications to these models and use the outputs directly in downstream tasks. While this flexibility is valuable, it creates what we call the *Oracle Fallacy*: treating an LLM prediction as a validated measure of a latent economic construct without first establishing whether its errors permit valid statistical inference. Benchmark performance alone is not sufficient, because domain validity depends on how model errors relate to the underlying construct and to other variables used in the empirical design.

We make three contributions. First, we survey the recent finance literature using LLMs and classify studies by the strength of their validation procedures. Second, we provide a full validation walkthrough in an M&A setting and document two important failure modes. The first is *minority class collapse*, in which high aggregate accuracy conceals near-zero recall for economically important rare categories. The second is *ground truth crisis*, in which low agreement among domain experts makes conventional accuracy metrics difficult to interpret. Third, we propose a five-step validation protocol, including a novel *Reasoning Audit* that compares an LLM's rationale with expert annotation notes using independent evaluators. The protocol is designed to distinguish label disagreement from reasoning quality and to provide a more credible validity test for subjective financial constructs.

2 What Drives Corporate M&A Deals? (with Debarshi Nandy and Shubhranshu Shekhar)

Past literatures have investigated a lot in the valuation created by M&A deals, and it is found that the announcement effects vary distinctly by acquirors' size, with smaller firms generally receive higher CARs, and large firms suffers massive losses around the announcement of the deal. Also, the long-term values created by the deals is negative. These outcomes are not favored by investors and shareholders who initially vote for the merging. We want to find whether the reasons of doing the M&A provides new insights about the deal valuations and explains the missing relationship between expectations and realizations of the value created.

In order to classify deals based on the rationale of the M&A deals, we follow the steps proposed in the paper *The LLM Oracle Fallacy in Finance* by using LLMs on the deals' press release documents. We classify acquisitions into four primary motivations—Efficiency, Finance, Growth, and Technology. The determinants of deal type suggest that acquirors enter different types of M&A transactions from systematically different financial positions. The market's initial assessment differs sharply across deal rationales. Finance and Growth deals receive the strongest positive market reactions at announcement, whereas Technology deals attract relatively weak market enthusiasm and downward analyst revisions. Yet these initial assessments do not map directly into long-run outcomes for all classifications. Abnormal returns deteriorate across most deal classifications as the post-merger horizon lengthens. Finance deals are the only category whose BHARs remain approximately flat and statistically indistinguishable from zero through the first two years, whereas Efficiency, Growth, and Technology deals already exhibit negative long-run performance over this period. By the three-year horizon, Efficiency, Finance, and Growth deals all show negative BHARs, while Technology deals are roughly flat and statistically insignificant. This pattern suggests that the rationale for an acquisition is informative not only about investors' initial expectations but also about how those expectations are realized over time.

Future Research

I have several projects that I am excited to pursue more fully after the job market and dissertation process. I expect my future research to continue building on my work in innovation and corporate finance, while becoming more interdisciplinary and expanding into related areas such as entrepreneurship.

As a natural extension of my dissertation, I plan to study the interaction between innovation and M&A by building on the datasets developed in my dissertation papers and the additional data infrastructure I have already constructed. One direction is to examine whether organic and acquired innovation have different effects on firm valuation and subsequent performance. Patent ownership often changes following an acquisition: some patents are transferred to the acquiror along with the target, while others may be sold to third parties around the transaction. These ownership changes provide an opportunity to study how individual patents are valued, how they fit within the acquiror's existing technological portfolio, and how acquired technologies contribute to future growth. A second direction is to extend my industry-level technological complexity measure to the

firm and patent levels. My job market paper shows that technological complexity is related to market capitalization and economic value at the industry level, but a more granular measure would allow for a much richer analysis of how complexity varies across firms and technologies.

For the first project, an important observation is that not all target patents remain with the acquiror after an acquisition. This suggests that acquirors may selectively retain technologies that complement their existing portfolios or support future strategic objectives. Studying this mechanism requires detailed patent ownership histories. The USPTO assignment database records changes in patent ownership, and I have already completed the matching of assignee names to Compustat firms. By combining these assignment histories with M&A transaction data, I can identify patents that are retained, transferred, or subsequently sold. This makes it possible to compare internally developed and acquired technologies and relate them to post-acquisition operating performance, deal valuation, and stock-market reactions. One hypothesis is that acquired technologies can complement internal innovation and improve subsequent operating performance when they fit well with the acquiror's existing technological capabilities.

The second project grows directly out of the analysis in my job market paper. Because the current technological complexity measure is constructed at the industry level, firm-level heterogeneity cannot be examined precisely. The next step is therefore to construct firm-level complexity measures using the CPC classifications of each firm's granted patents. This would broaden the range of questions that can be studied using the complexity framework and would also make it possible to compare public and private firms. Such a comparison is particularly interesting because private firms are often an important source of experimentation and frontier innovation, and firm-level measures would allow me to examine whether their technological portfolios differ systematically from those of public firms.